

# Pore space analysis - Hackathon-Polaron batches

**Answer:** On total pore space, mean image-area porosity trends upward Batch\_1 < Batch\_2 < Batch\_3 (Batch\_1: 0.088 [0.077-0.097]; Batch\_2: 0.099 [0.090-0.109]; Batch\_3: 0.108 [0.099-0.118]), consistent with the earlier rough cross-check (~0.089/0.100/0.109), and the ranking is stable under +/-10% threshold shifts - but it is only marginally supported statistically (Kruskal-Wallis  $p=0.054$  with all images,  $p=0.116$  without the two outlier images; the only nominally significant pairwise contrast, Batch\_1 vs Batch\_3 Holm  $p=0.039$ , does not survive outlier removal, Holm  $p=0.116$ ). On the silicon-relevant question, NONE of the pore measures next to the assumed-Si particles differs between batches (local 10-px ring porosity Batch\_1: 0.062 [0.051-0.075]; Batch\_2: 0.064 [0.050-0.078]; Batch\_3: 0.066 [0.060-0.072], KW  $p=0.872$ ; confined share Batch\_1: 0.439 [0.405-0.480]; Batch\_2: 0.471 [0.442-0.505]; Batch\_3: 0.439 [0.408-0.469],  $p=0.241$ ; boundary contact,  $p=0.801$ ). Pore space next to silicon is also 27-38% below the whole-image average in every batch (local/whole ratio 0.62-0.73). Interpreted against the paper - 40% bulk-porosity anodes outlived 30% ones (347 vs 225 cycles to 75% capacity) and too little porosity promoted Li plating - the batch with least porosity would nominally be at greatest risk, but see the risk section: no ranking is supported. Assumption: bright particles are silicon-based, grey bulk is graphite, from image appearance only.

*Assumption throughout: bright BSE particles are taken to be silicon-based and the grey bulk to be graphite, from image appearance only (no EDS confirmation).*

## Method

BSE images only (31 total: Batch\_1=7, Batch\_2=7, Batch\_3=17). Grey levels are the per-pixel median of the three RGB channels; no information banner was present. Each image was normalised to its 1st-99.5th percentile range and Gaussian-smoothed (sigma 1.5 px). Pores: below the LOWER cut of a 3-class multi-Otsu; objects <20 px removed. Bright particles: histogram-valley threshold (upper multi-Otsu cut as flagged fallback), opening disk  $r=2$ , objects <150 px removed, holes filled, watershed-split - identical to the silicon\_size analysis. Bright pixels win where masks overlap; labels are pore/bulk/bright. Pore classes: large gap >0.5% of image area, crack/slit if major:minor >3, else compact. Pore size = mean local thickness (2 x mean distance transform). Chords are horizontal/vertical pore run lengths. Depth profile = pore fraction in 10 horizontal bands. Around each interior particle: pore fraction of the 5/10/20-px annulus (nearest-particle attribution via distance transform), share of boundary within 3 px of a pore, and 'confined' = zero pore in the ring. Pixel size ~25.0 nm/px in every image (TIFF tags). The IMAGE is the independent unit: Kruskal-Wallis + pairwise Mann-Whitney (Holm) + bootstrap CIs resampling images. Threshold sensitivity: pore threshold shifted +/-10% per image.

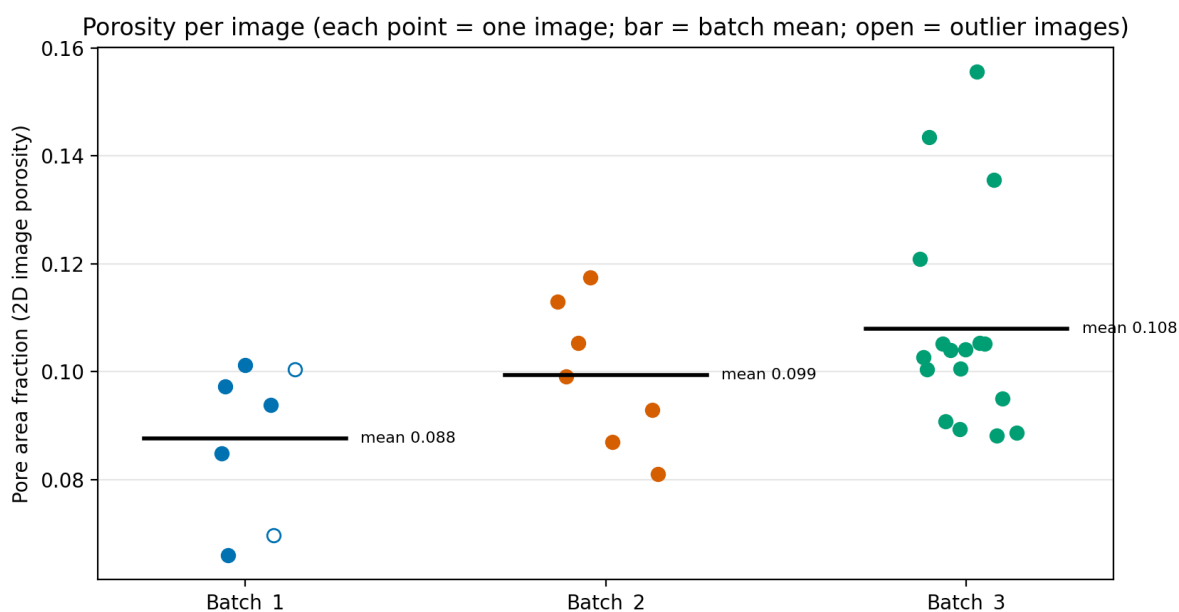


Fig. 1 - Image-area porosity per image, grouped by batch; horizontal bars are batch means; open markers are the two low-contrast outlier images.

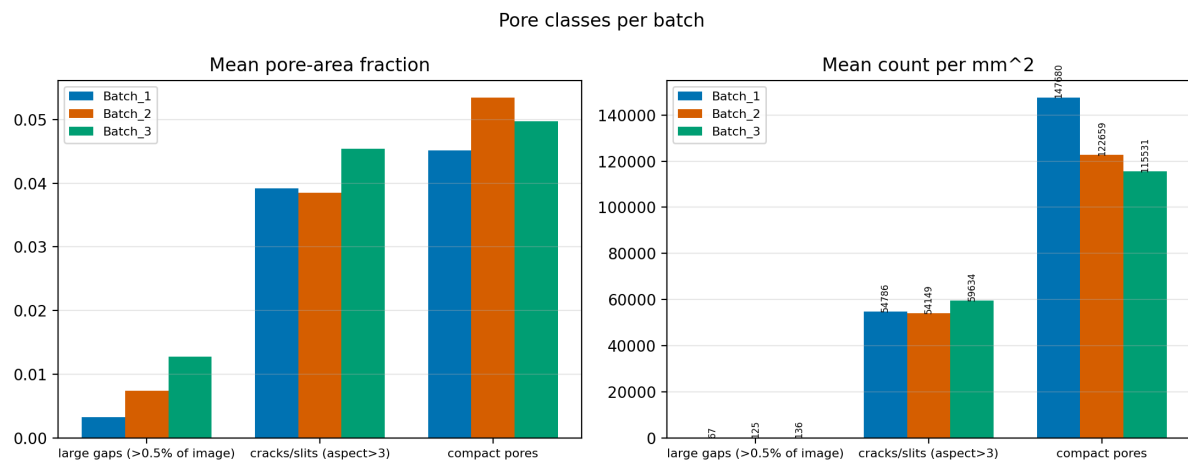


Fig. 2 - Mean area fraction (left) and areal density (right) of the three pore classes per batch.

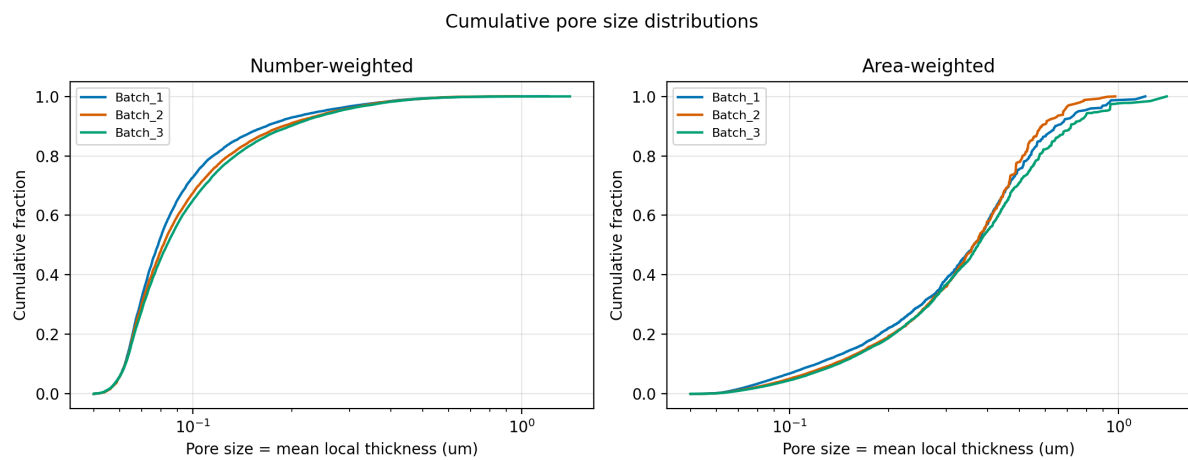


Fig. 3 - Cumulative pore-size (mean local thickness) distributions, number- and area-weighted, per batch.

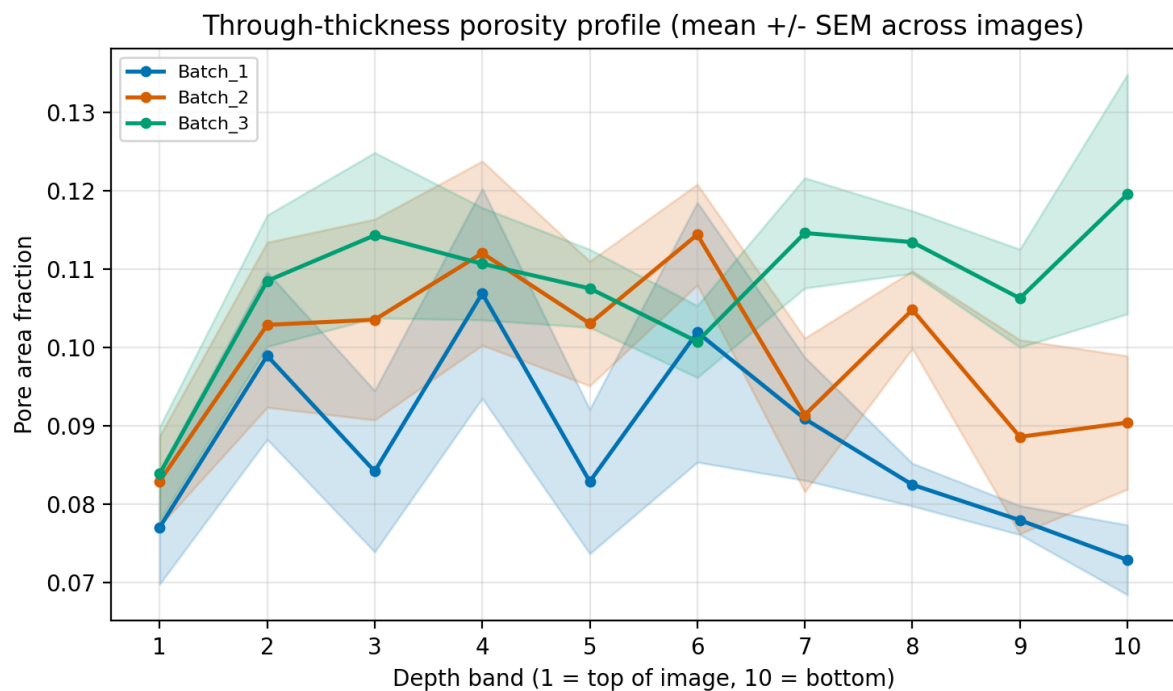


Fig. 4 - Porosity in 10 horizontal bands from top to bottom of each image, batch mean +/- SEM.

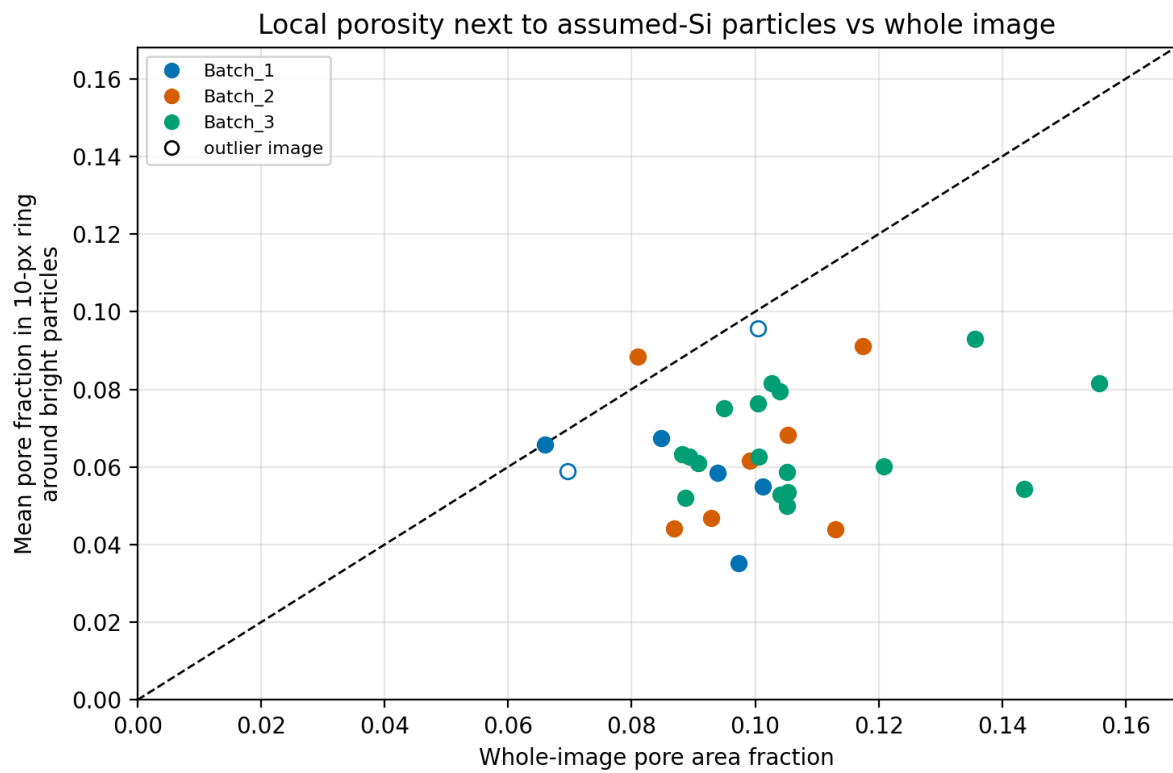


Fig. 5 - Mean pore fraction in the 10-px ring around assumed-Si particles vs whole-image porosity, per image; dashed line is equality.

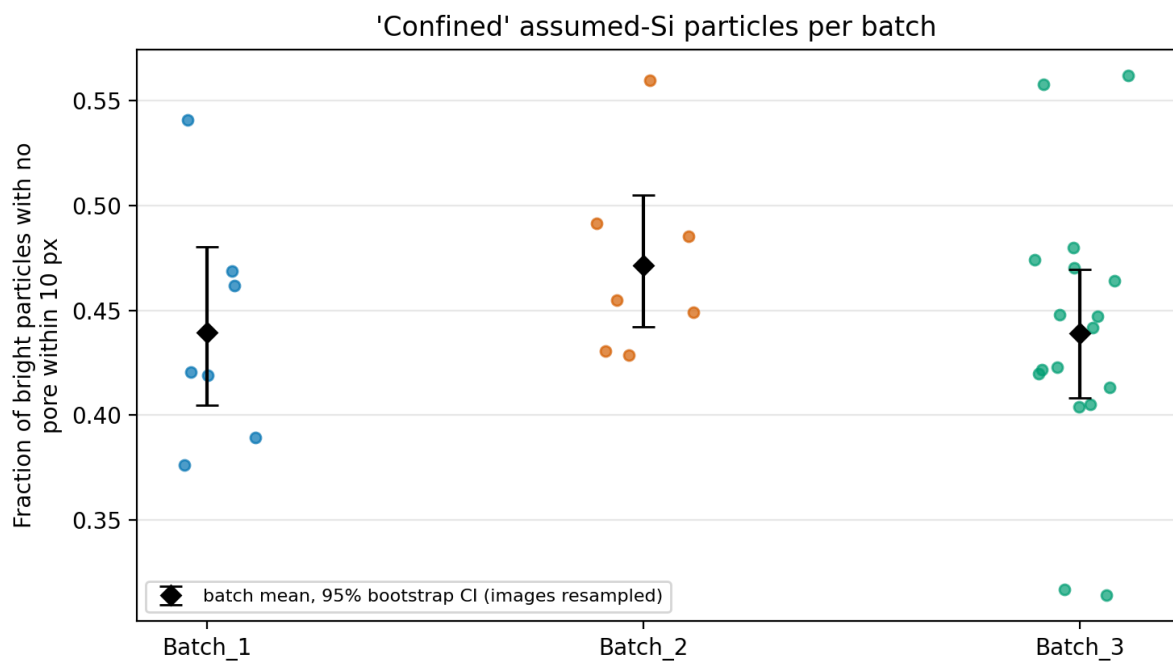


Fig. 6 - Fraction of assumed-Si particles with no pore within 10 px, per image and batch mean with 95% bootstrap CI.

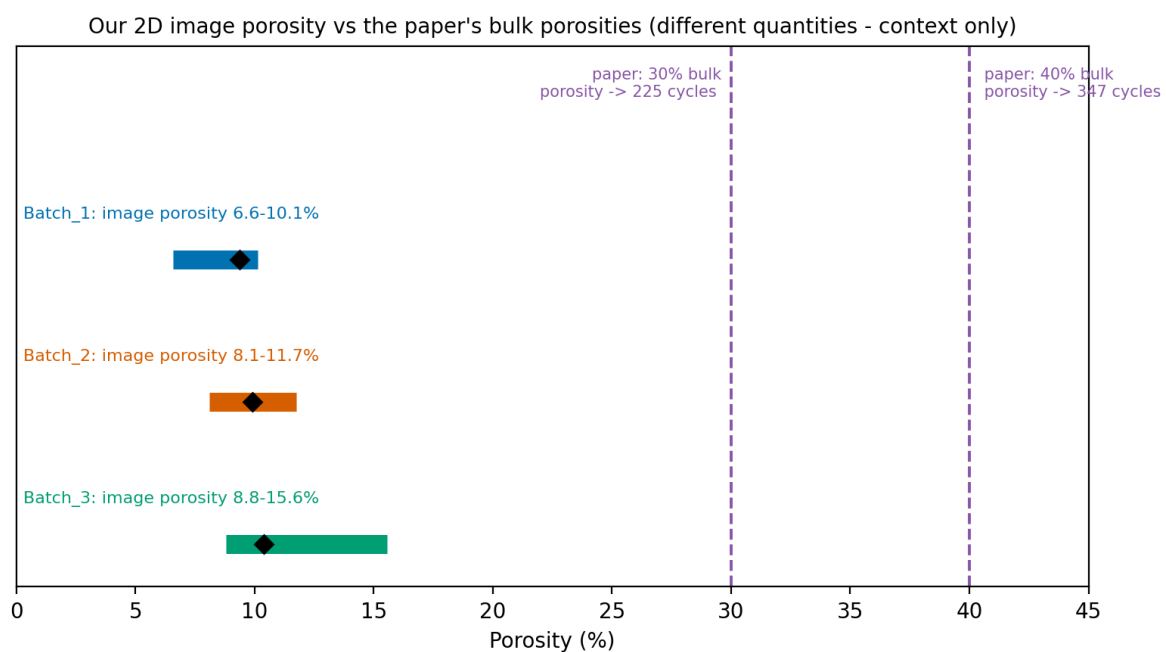


Fig. 7 - Our 2D image porosity per batch (bars: min-max range, diamond: median) next to the paper's two bulk porosities and their cycle lives - different quantities, shown for context only.

## Batch statistics

| batch   | images | porosity | 95% CI        | local poro 10px | 95% CI        | confined | 95% CI        | pore D50 um | chord h/v |
|---------|--------|----------|---------------|-----------------|---------------|----------|---------------|-------------|-----------|
| Batch_1 | 7      | 0.088    | [0.077,0.097] | 0.062           | [0.051,0.075] | 0.439    | [0.405,0.480] | 0.08        | 1.18      |
| Batch_2 | 7      | 0.099    | [0.090,0.109] | 0.064           | [0.050,0.078] | 0.471    | [0.442,0.505] | 0.08        | 1.17      |
| Batch_3 | 17     | 0.108    | [0.099,0.118] | 0.066           | [0.060,0.072] | 0.439    | [0.408,0.469] | 0.08        | 1.18      |

## What the paper says, and whether it applies

The paper (Profatilova et al., ACS Appl. Energy Mater. 2020, 3, 11873-11885, DOI 10.1021/acsaem.0c01999) tested Si/C/graphite composite anodes in 18650 cells at two anode porosities. Verbatim from the abstract: "Anode porosity variation between 30 and 40% led to a significant difference in the cycle life of 18650 cells: 225 and 347 cycles were achieved, respectively, before the drop of a discharge capacity to 75% of the initial value" and "Li+ ion loss in the solid electrolyte interphase layer was the main reason for the performance loss in the case of 40% anode porosity", while "Compression of the anode to 30% of porosity resulted in Li metal deposition on the negative electrode surface and within the separator during cycling ... Li+ ion transport was largely impeded by this deposit leading to poor C-rate performance." IMPORTANT ACCESS NOTE: the full text is paywalled and no open-access copy could be obtained (checked ACS, OpenAlex, Unpaywall, HAL, figshare). The figures/quotes above come from the publisher-hosted abstract and the free Supporting Information (Table S1 shows thickness measured by gauge and by cross-SEM). The paper's porosity is the BULK value set at cell build (thickness/mass based), a different quantity from our 2D image-area porosity: our ~10.1% area fractions and the paper's 30%/40% bulk values cannot be compared numerically; only the qualitative direction - over the studied range, more pore space next to/within the anode correlated with LONGER cycle life, while too little porosity promoted Li plating - is invoked. Whether the relationship is monotonic or has an optimum cannot be told from two points.

## Risk ranking

No defensible risk ranking is supported. On the paper's direction alone - more porosity gave longer cycle life over its 30-40% bulk range, and too little porosity promoted Li plating - the descriptive ordering would nominally put Batch\_1 (lowest mean image porosity) at greatest risk. But the total-porosity difference is at/below the noise floor (Kruskal-Wallis  $p=0.054$ , not robust to outlier removal), and the measures that matter most for silicon - pore space in the ring around each bright particle, boundary contact, and the share of confined particles - show no batch difference at all (KW  $p=0.872/0.241/0.801$ ). Any ranking would therefore rest on a non-robust ~2-point porosity difference in a quantity (2D area fraction) that is not even the same measurement the paper varied (bulk calendered porosity).

## Limitations

- Material identity is assumed: bright BSE particles = silicon-based, grey bulk = graphite, from image appearance only (no EDS).
- 2D image porosity (area fraction on a polished section) is not the paper's bulk porosity (thickness/mass); only the direction of the paper's trend is used.
- The paper's full text was not accessible (paywalled; no OA copy). Quotes are from the abstract and SI only; no detail beyond them is relied on.
- Pore checks: the dark class is essentially black - median 78% of pore pixels are below grey level 30/255, so pores read as empty voids rather than resin-filled shadowed solid. Bright material sitting inside voids (e.g. back walls) is essentially absent: only 2 of 31 images contain any (max 1 object), so bias to porosity is negligible; where present those pixels are classified bright, keeping them out of the pore count.
- Threshold sensitivity: The batch ranking on mean porosity is stable under +/-10% pore-threshold shifts.
- Pixel size (~25.0 nm/px) comes from TIFF resolution tags, not a measured scale bar; a systematic error would shift all lengths together (porosity itself is dimensionless and unaffected).
- The image is the independent unit ( $n = 7, 7, 17$ ); Batch\_3's larger count gives pairwise tests involving it more power.
- Two Batch\_1 outlier images (4ih2ggld, 5n1q8atc) at lower contrast; all statistics are reported with and without them.
- Ring porosity attributes each non-bright pixel to its nearest particle; in dense clusters the ring contains less free space by construction.
- Cross-check: earlier rough porosities were ~0.089/0.100/0.109 for Batch\_1/2/3 ( $p=0.07$ ). Our means are Batch\_1=0.088; Batch\_2=0.099; Batch\_3=0.108. Differences vs the rough pass come from the same-method multi-Otsu pore cut plus <20 px cleaning and the bright-wins overlap rule.